App feel

Summary

Make GatherPack feel like an app: pages that change smoothly, saves that don't lose your place, an icon on the home screen, and reminders that reach a phone. This stays on the Hotwire stack GatherPack already uses (Turbo and Stimulus), which is built for this and is barely used for it today. No screens are rewritten, and no separate front end is needed.

What it might look like

This is a concept mockup with sample data. The phone screens are drawn, not captured from real devices, and they aren't a final design. Smoother page changes can't be shown in a still image, so this covers only the phone features: a reminder on the lock screen, the app on the home screen, and the settings for choosing which reminders to get.



Problem

Pages can flicker or jump as they change, which makes a site feel old even when it is fast. Timed page visits on a desktop with a good connection took about 190–380 ms with almost no layout shift, so page speed alone doesn't explain it. Saving forms and phones weren't part of that test. The code points to these likely causes:

- **The whole page is replaced on every click and save.** No page uses Turbo's morphing, and nothing stays in place between pages, including the sidebar.

- **Some actions do a real full reload.** There are 19 Turbo opt-outs across 14 files, including the dashboard, calendar, time clock and sidebar. Some are needed, such as sign-in and external login providers. Others may not be.
- **The dashboard appears twice.** On a return visit it draws a cached copy, then replaces it about half a second later.
- **Dashboard pieces arrive late.** The calendar and widgets load after the page, with no space held for them.
- **The first load is heavy.** About 70 files and 4.3 MB uncompressed, including all of Font Awesome and the web fonts from outside CDNs.

GatherPack is also only partly a PWA. It has a manifest that opens without the browser bar, but its short name is blank, and its service worker is entirely commented out and never registered. It has no way to send a notification to a phone. Reminders go by email, where they are easy to miss.

What it does

- **Saves that keep your place.** Saving on a page and returning to it updates only what changed and keeps the scroll position.
- **A steady frame.** The header and sidebar stay still as pages change. Dashboard cards hold their space while they load.
- **No double draw.** Pages with live information don't show a stale copy first.
- **Faster first and repeat visits.** Fonts and icons are served by GatherPack and cached.
- **Install to the home screen.** A proper name and icon, and it opens full screen.
- **Phone notifications.** People can opt in to get reminders on their phones, such as a form due or a shift tomorrow, alongside email. iPhones allow this only after the app is installed, so the page explains how.

Offline use is deliberately left out. Cached pages could leave student and family information on shared devices, and could keep showing things after someone's access has changed. Without a connection, the app shows a plain "you're offline" page.

How it's built

- **Smoothness.** `turbo-rails` 2.0 already includes Turbo 8. Turn on morph and scroll preservation for same-page refreshes, mark the sidebar `data-turbo-permanent` with a small Stimulus controller to keep its active link correct, use `no-preview` on pages with live data, and add placeholders for late-loading cards. Review each Turbo opt-out and remove the ones no longer needed.
- **Assets.** Host Font Awesome and the fonts locally. Trimming icons isn't simple, because admins choose icons by name, so the full set stays and relies on caching.
- **Installable app.** Fill in the manifest, and register a service worker that caches only versioned CSS, JavaScript, fonts and icons, plus the offline page. It never caches signed-in pages. Turning the flag off also unregisters it, so a bad version can't get stuck on devices.
- **Notifications.** A new table stores each person's device subscriptions, using standard Web Push with no outside account required. Existing reminders, such as `FormScheduleJob` and `SignupReminderJob`, add it as a delivery channel. Messages carry a short title and a link, with no personal details on the lock screen. Permission is requested only after someone chooses to turn it on.

Other options, if a need appears:

- **A small JavaScript component on one screen**, for something like a drag-and-drop board. GatherPack already does this with FullCalendar and CodeMirror.
- **Hotwire Native**, which wraps these same pages in App Store apps. This makes sense only if Web Push proves inadequate, since it adds store accounts and reviews.
- **A separate front end** is the only option that means rewriting screens. It would be justified only by full offline use or a highly interactive app throughout, and GatherPack needs neither.

Size: Small to medium, in independent parts. Smoothness and assets need no migrations and touch a few layouts and templates. Notifications need one migration, one gem and new keys in server configuration. Flags off by default.

Risks

- **Scripts that assume a fresh page.** Morphing and a permanent sidebar can disturb Stimulus controllers, Bootstrap menus and modals, and the calendar. Each needs testing, and morphing can be turned off per page.
- **Removing an opt-out that exists for a reason.** Each may be working around something undocumented, so remove them one at a time.
- **A stale service worker.** Caching mistakes are hard to undo on devices already running it. Keep what it caches small, and test the off switch before shipping.
- **Shared devices.** The kiosk stays out of the installable app and notifications, and signing out removes that device's subscription.
- **Notification fatigue.** Too many messages get them turned off, so people choose which reminders they want.

First step

Reproduce the flicker before fixing it. Record the main tasks on a real phone and on a desktop, including saving forms, and list exactly what flickers. Then apply the smoothness changes one at a time and record them again. Install and notifications follow as separate pieces of work.

Adoption

Independent of the other features in this folder, and could ship first. The smoothness and asset changes fix issues in core and are good candidates for upstream. Notifications need a generic delivery channel that other features can use, so reminders aren't each wired to it separately.